

Difference Between Parallel Systems and Distributed Systems

(summarized by ChatGPT)

Parallel systems and distributed systems both involve multiple processors working together, but they have fundamental differences in architecture, memory management, and use cases.

◆ Parallel System

- **Definition:** A system where multiple processors work together within a single machine, sharing memory and executing computations simultaneously.
 - **Structure:** Multiple processors are physically connected within a single system (shared memory architecture).
 - **Characteristics:**
 - Primarily used for **high-performance computing (HPC)** and **scientific calculations**.
 - All processors share the **same operating system (OS)** and memory space.
 - **Fast data sharing**, but synchronization issues such as **race conditions** and **deadlocks** may occur.
 - **Examples:**
 - Multi-core processors (CPU with multiple cores).
 - Supercomputers with shared memory-based clusters.
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◆ Distributed System

- **Definition:** A system where multiple independent computers (nodes) are connected via a network and work together as a single system.
- **Structure:** Independent computers (nodes) communicate through a network (e.g., LAN, WAN, the Internet).
- **Characteristics:**
 - Each node has its **own memory and CPU** and operates independently.
 - Work and data are **distributed** among nodes (load balancing).
 - **Highly scalable**, as nodes can be easily added.
 - **Fault tolerance** is high—if one node fails, the system can continue running.
 - However, **network latency** and **data consistency** issues may arise.
- **Examples:**
 - **Cloud computing** (Amazon AWS, Google Cloud, Microsoft Azure).
 - **Distributed databases** (Google Spanner, Apache Cassandra).
 - **Blockchain systems** (Bitcoin, Ethereum).
 - **Microservices architecture**.

◆ Comparison: Parallel System vs. Distributed System

Feature	Parallel System	Distributed System
Hardware Structure	Multiple processors share the same memory	Independent computers (nodes) connected via a network
Memory Architecture	Shared memory	Distributed memory
Operating System	A single OS is shared by all processors	Each node has its own OS
Processing Method	A single task is divided among multiple processors	Different tasks are distributed across nodes
Data Sharing Speed	Fast (direct memory access)	Slower (network communication required)
Scalability	Limited (hardware upgrades needed)	High (easily scalable by adding nodes)
Fault Tolerance	Low (failure of a processor can impact the entire system)	High (system can continue working even if some nodes fail)
Typical Use Cases	Supercomputers, AI computing, scientific simulations	Cloud computing, blockchain, distributed databases

✓ Summary

- **Parallel systems** involve multiple processors **sharing memory** within a single machine, focusing on **performance improvement**.
- **Distributed systems** consist of multiple independent computers connected via a **network**, emphasizing **scalability and reliability**.
- **Parallel systems are performance-oriented**, while **distributed systems are designed for scalability and fault tolerance**.